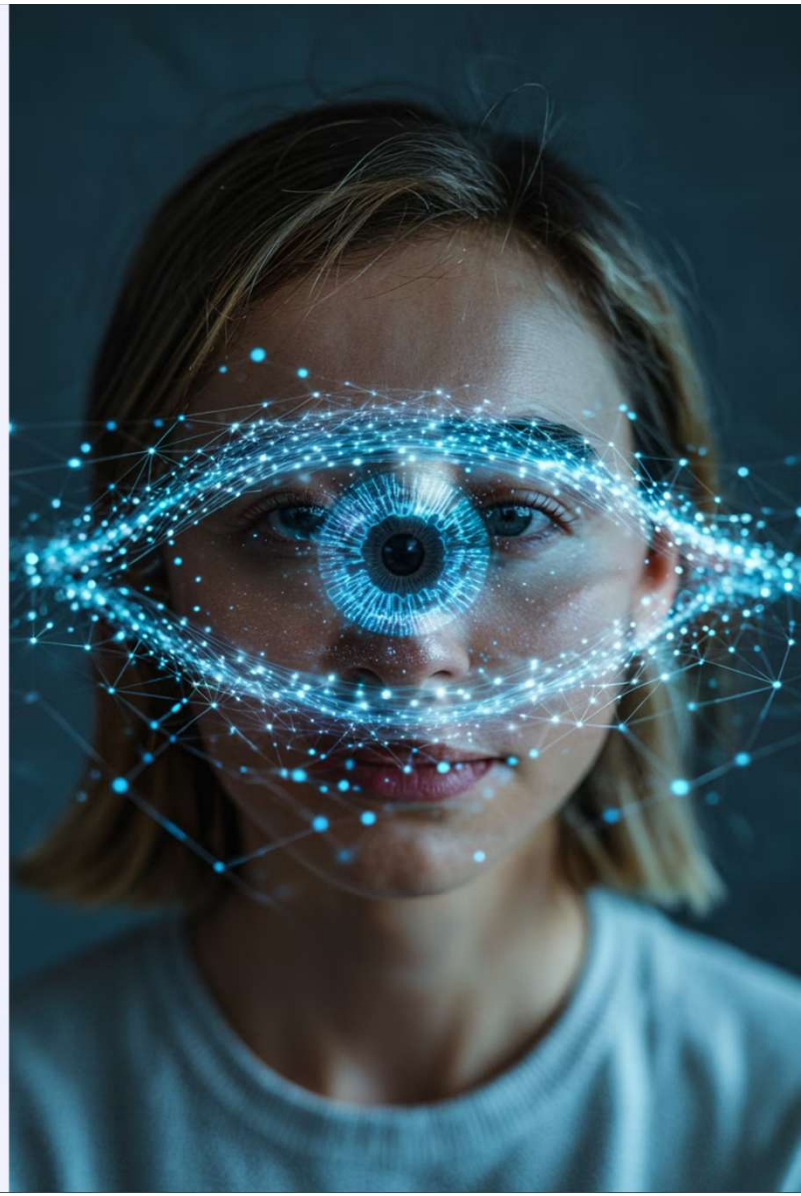
Introduction to Computer Vision Concepts

A foundational module exploring how machines learn to see, interpret, and generate visual content.



MODULE OVERVIEW

What You'll Learn

01

Identify Computer Vision Tasks

Classify the core tasks in computer vision.

02

Understand Image Filters

Explain how filters extract features from images.

03

Explore CNNs

Describe the architecture of Convolutional Neural Networks.

04

Discover Vision Transformers

Understand how ViTs use attention for visual data.

05

Generative AI for Images

Explain how AI creates images from text prompts.

What Is Computer Vision?

Computer Vision is a domain of AI that enables machines to **interpret and process visual data** — images and videos — simulating human vision. Computers analyze visual input from cameras or digital media to make intelligent decisions.

🚙 Autonomous Vehicles

Detecting traffic, pedestrians, and road conditions in real time.

🛒 Smart Retail

Automated checkouts that identify products without barcodes.

🚪 Smart Security

Doorbell cameras that recognize familiar visitors and alert homeowners.

Computer Vision Tasks & Techniques

Computer Vision uses AI techniques to process images, videos, and live streams — extracting meaningful information at varying levels of detail.

Image Classification

Assigns a single label to an entire image (e.g., "apple" or "banana"). Used in smart checkout systems.

Object Detection

Identifies multiple objects and their locations using bounding boxes. Detects several items in one image.

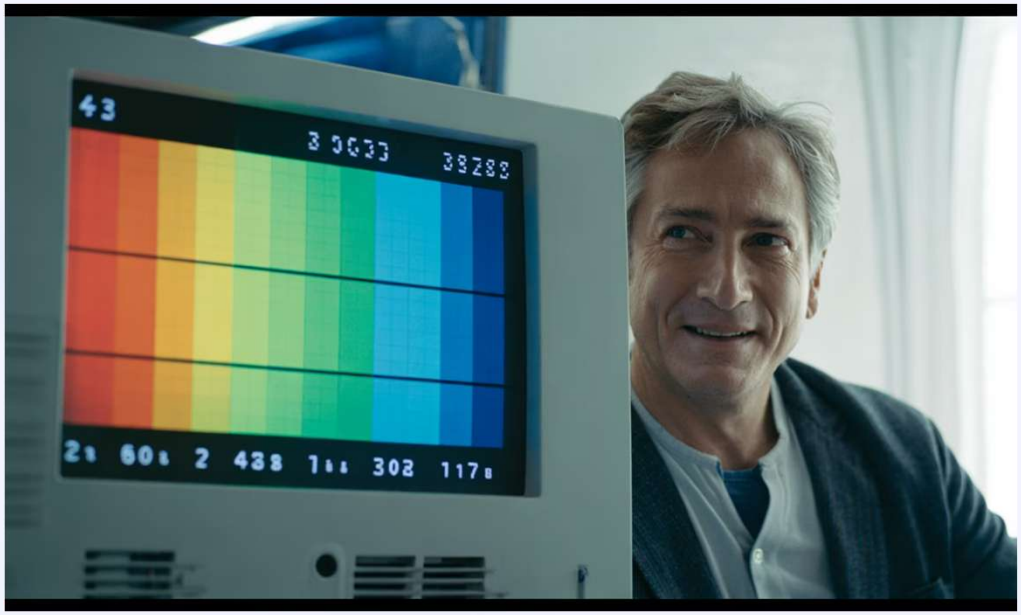
Semantic Segmentation

Classifies each pixel for precise object identification, providing detailed object boundaries.

Contextual Image Analysis

Understands relationships between objects and activities to generate captions like "Person eating an apple."

Images & Image Processing



How Computers See Images

An image is a **matrix of pixel values** ranging from **0 (black) to 255 (white)**.

- **Grayscale:** Single 2D array of pixel intensities
- **RGB Color:** Three channels — Red, Green, Blue — combined to produce all colors

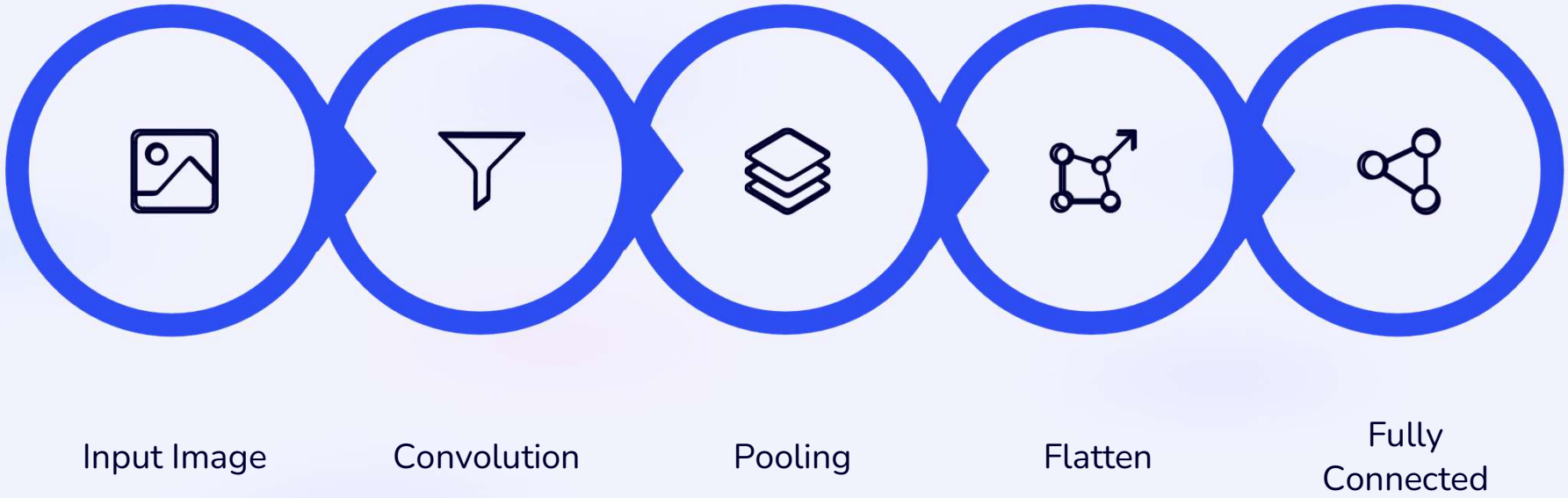
Filters & Convolution

Filters (kernels) scan the image using **convolution**, producing new pixel values through weighted calculations.

Example: The **Laplace filter** highlights object boundaries through edge detection.

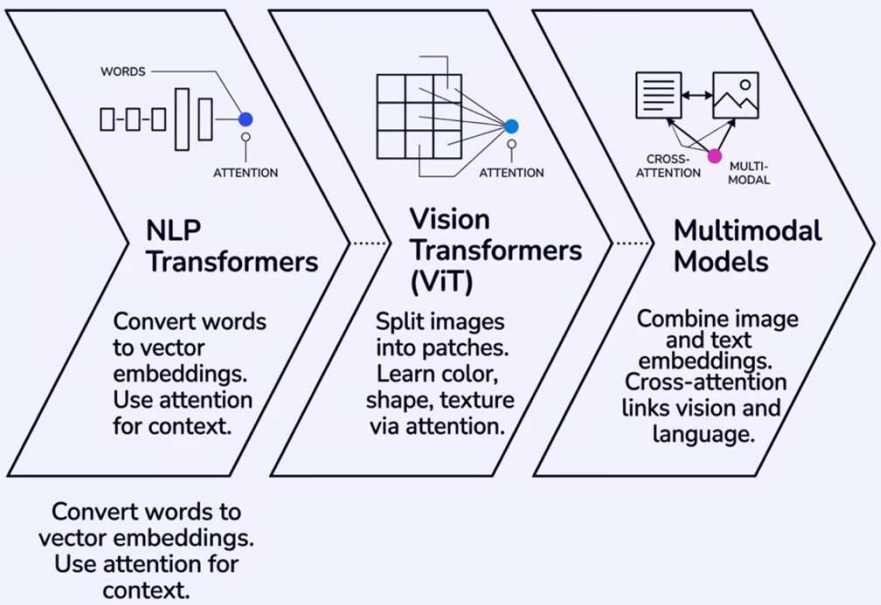
Convolutional Neural Networks (CNNs)

CNNs are deep learning models that **automatically learn visual features** from images — no manual feature engineering required.



During training, the network starts with random weights, compares predictions to actual labels, and adjusts weights over multiple epochs to minimize error — progressively improving accuracy.

Vision Transformers & Multimodal Models



Beyond CNNs

Transformers, originally designed for language, now power modern vision systems. **Vision Transformers (ViT)** split images into patches and apply the **attention mechanism** to learn global visual relationships.

Multimodal AI

By combining **image and text embeddings** through cross-attention, multimodal models can caption images, answer visual questions, and interpret scenes with language-level understanding.

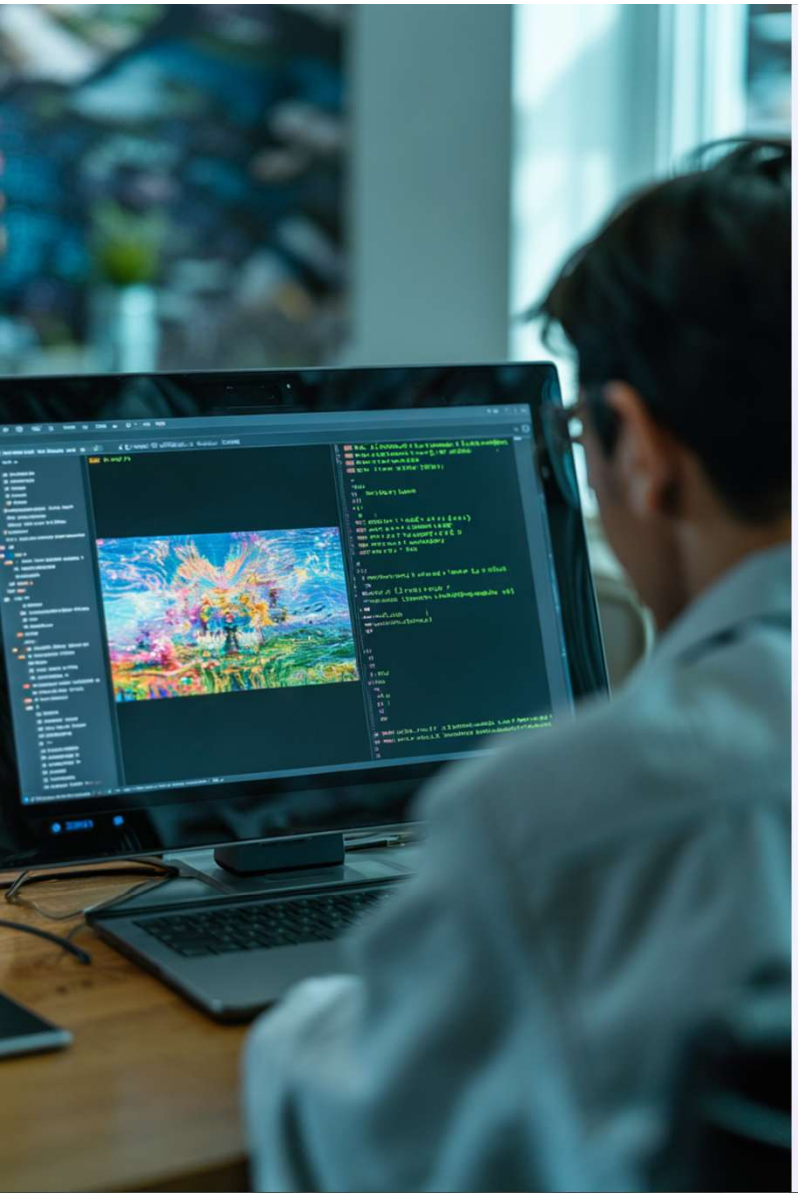


Image Generation & Key Takeaways

Text-to-Image Generation

Multimodal models translate a text prompt (e.g., *"A dog carrying a stick"*) into visual features, generating images step-by-step.

Diffusion Technique

Starts with **random noise** and gradually refines it through iterations, comparing output to the prompt until a realistic image emerges.

Video Generation

Advanced models also generate videos, maintaining **motion consistency** and real-world physical behavior across frames.

Key Insight: Computer vision has evolved from simple pixel analysis to AI systems that **understand, interpret, and generate** visual content.